AI lets one person do a team's work.

Fewer jobs, or more companies?

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I've used AI in my work every day for four or five years. Around December 2025, it crossed a line for me. I could suddenly do work I never had time for before.

That got me wondering whether one person doing a team's work means fewer jobs, or more companies. I tested it with a small model of an economy, Census data on new businesses and the newest research. What I learned: replacement isn't the only possible outcome. New companies can make up for lost jobs, but only under certain conditions, and we can watch for those conditions in real data.

How it started

The first thing I did with my new free time was start a project that had been rattling around in my head for almost a decade: a Rust networking library built on DPDK. My plan was to hand-write a UDP layer. I ended up with a TCP stack that runs in userspace, the kind of thing that usually takes a team of specialists years. I made 841 public GitHub contributions in the last twelve months, against 65 in all the years before that combined.

The effort it takes to do more has dropped, and I feel it. Most of what I read about this is about **replacement**: companies adopt AI and need fewer people. I wanted to know if that's the whole story.

Two forces inside every company that adopts AI

When a company adopts AI, two things happen at once, and they pull in opposite directions.

1. **AI replaces some of the work.** Tasks that took a person now take a prompt.
2. **AI makes the work cheaper.** Cheaper output means the company can charge less, sell more and take on things it couldn't afford before.

If the second force wins, the company grows and hires, even though AI is doing part of the job. If the first force wins, it does the same work with fewer people. Whether whole companies behave like the second case is the big open question, and nobody has pinned it down yet.

Starting a company gets cheaper too

AI also lowers the cost of starting a new company. We've seen a version of this before. Cloud computing made starting a software company dramatically cheaper. From 2006 to 2019, applications for new businesses rose **34%**, while applications from people who planned to **pay wages** fell **48%** (Census Business Formation Statistics). That doesn't prove cloud caused either trend. But it's a warning: more new businesses doesn't automatically mean more new employers.

I built a toy economy to see what decides it

I built a simple simulated economy of about 175,000 workers. Companies are born, grow if they're productive, shrink if they're not, and close when they stop making money. Then I add AI. Nobody knows how strong AI's effects are, so I ran the economy under many different guesses, each with a twin where AI never arrives.

My first results were too rosy. In most of the good runs, the big companies never cut jobs at all. When I looked only at runs where big companies did cut, the picture got much more mixed, and when I let the price of AI vary too, it got worse again. The answers moved a lot with my assumptions, so I'm not going to give you odds. I'll give you the conditions instead.

Three questions decide it:

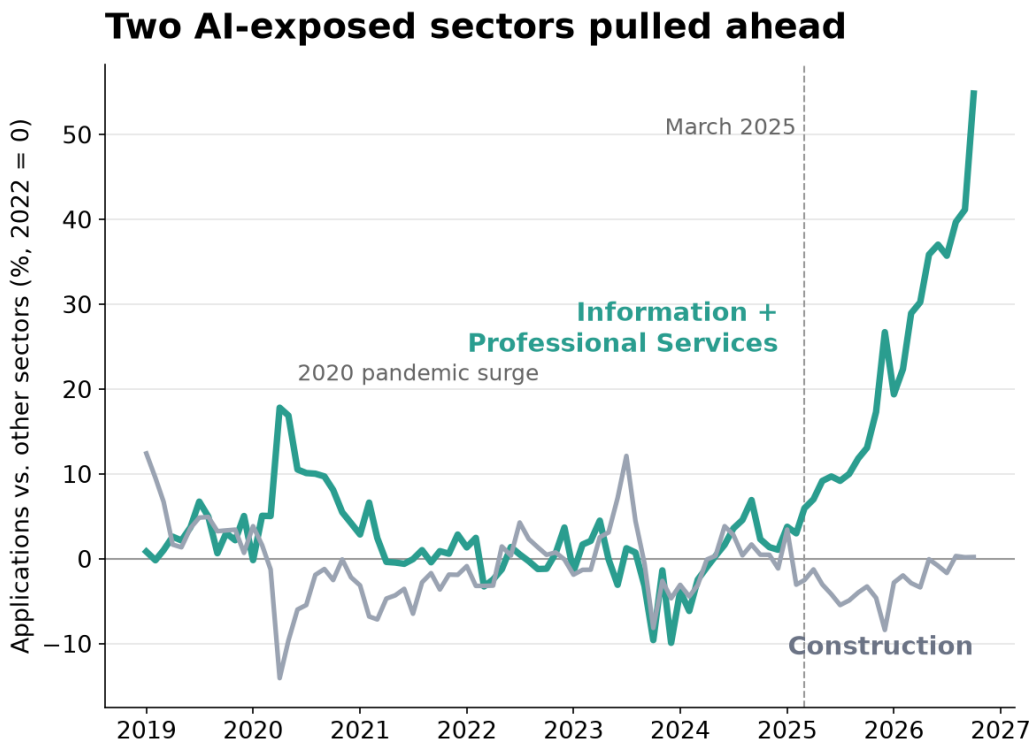
1. **Do companies that adopt AI grow, or shrink?**
2. **Are AI-native startups better, and do they hire?** In my runs, making startups cheaper did little on its own. What mattered was whether the new companies were more productive than the startups they replaced.
3. **Is the pipeline of founders still full?**

In my model, new companies reliably made up the losses when companies adopting AI grew, or when AI-native startups were much more productive than the ones before them. They reliably failed to when AI replaced people easily *and* the new startups were no better. In between, the results were mixed.

What the data shows so far

Lots of new businesses, very few new employers. Business applications are at a record, up **10.9%** from January to October 2026 compared with the same months of 2025. Applications from people planning to pay wages are *down* **16.5%**. Only **6.8%** of this year's applications plan to pay wages, and just **4.7%** in the Information sector (Census BFS, November 2026 release).

Two of the most AI-exposed industries pulled ahead. Starting around March 2025, applications in Information and Professional Services pulled away from the rest of the economy. On average in 2026 they ran roughly a third higher, relative to other sectors, than in 2022. Most of that came from applications the Census Bureau rates as unlikely to hire.



Source: Census Business Formation Statistics, monthly, through Oct 2026.

Not every exposed industry followed: Finance is also highly exposed to AI, and its applications fell relative to other sectors. And there's a less hopeful reading of the surge: laid-off tech workers registering as contractors. This data can't tell the two stories apart.

Few companies are cutting because of AI, yet. In the Census Bureau's Business Trends and Outlook Survey (November 2025 to January 2026), about **2% of businesses using AI** said they cut staff because of it, and companies using AI more deeply were more likely to see employment fall (Bonney et al. 2026). **AI-native startups are lean:** 12–25% smaller than comparable startups (Kim and Koning 2026).

Other researchers are finding the same tension

- **Bena, Bian and Giannetti** find that industries more exposed to generative AI saw **20% more startups**. Each one is smaller, but together they added jobs and raised pay among new firms (Prompted to Start, July 2026).
- **Gupta, Qian, Simintzi and Sun** find that startups exposed to AI cut jobs, mostly junior roles. About half of the workers who left took around six months to find new work, and many took lower-paying jobs. New startups made up the difference, so total startup employment held steady (Generative AI and Entrepreneurship, April 2026).

The solo founder bet

The surge in what look like solo ventures didn't surprise me. My bet is that many of those founders will hire once they make enough money. The odds I'm betting against are real: the Census Bureau

projects that the roughly 500,000 business applications filed in October 2026 will produce about 28,000 new employers within a year, about one in eighteen.

What this doesn't show

- **My model holds wages fixed**, so it can say which way things lean but not how many jobs.
- **It's a rough sketch, not a carefully calibrated model.** The scenarios show what could happen under different assumptions, not what will happen.
- **Where the good outcomes end depends on a number I had to assume**: how strongly a company's sales respond when its costs fall.
- **The Census data is descriptive.** It shows when things changed, not why.
- **It counts jobs, not good jobs.**

So, fewer jobs or more companies?

More new businesses, almost certainly. Whether they add up to as many jobs is still open. It depends on whether companies that adopt AI grow, and on whether this wave of new, smaller companies starts hiring. I don't think replacement is the only possible future, but the other one needs people willing to start small companies and willing to work at them.

Signal	Where to find it	Good sign
Jobs at companies that adopt AI	Census Business Trends and Outlook Survey	The 2% stays low as AI spreads
Hiring at AI-native startups	Studies like Kim and Koning's	Headcount grows with revenue
New employer businesses	Census Business Formation Statistics	Planned-wage applications start rising

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The model, the Census data and the code behind the charts are at https://github.com/gspivey/ai-employment-offset. This is a personal project, and the views here are my own, not my employer's. I used AI tools to help build the model and edit this summary.